



Database Management Systems

Converting E-R Diagram to Relational Schema

Anchor: Dr. Nagasundari S, Dept of CSE

Database Management Systems

Unit 1: Introduction to Database Management and SQL

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Dr. Nagasundari S

Department of Computer Science and Engineering



- The relational model represents the database as a **collection of relations**. Informally, each relation resembles a **table of values**.
- When a relation is thought of as a table of values, **each row** in the table represents a **collection of related data values**.
- The table name and column names are used to help to interpret the meaning of the values in each row.
- In the relational model terminology, a **row** is called a **tuple**, a **column** is called an **attribute**, and the **table** is called a **relation**.

- Informally, a **relation** looks like a **table** of values.
- A relation typically contains a **set of rows**.
- The data elements in each **row** represent certain facts that correspond to a real-world **entity** or **relationship**
- In the formal model, rows are called **tuples**
- Each **column** has a column header that gives an indication of the meaning of the data items in that column
- In the formal model, the column header is called an **attribute name** (or just **attribute**)

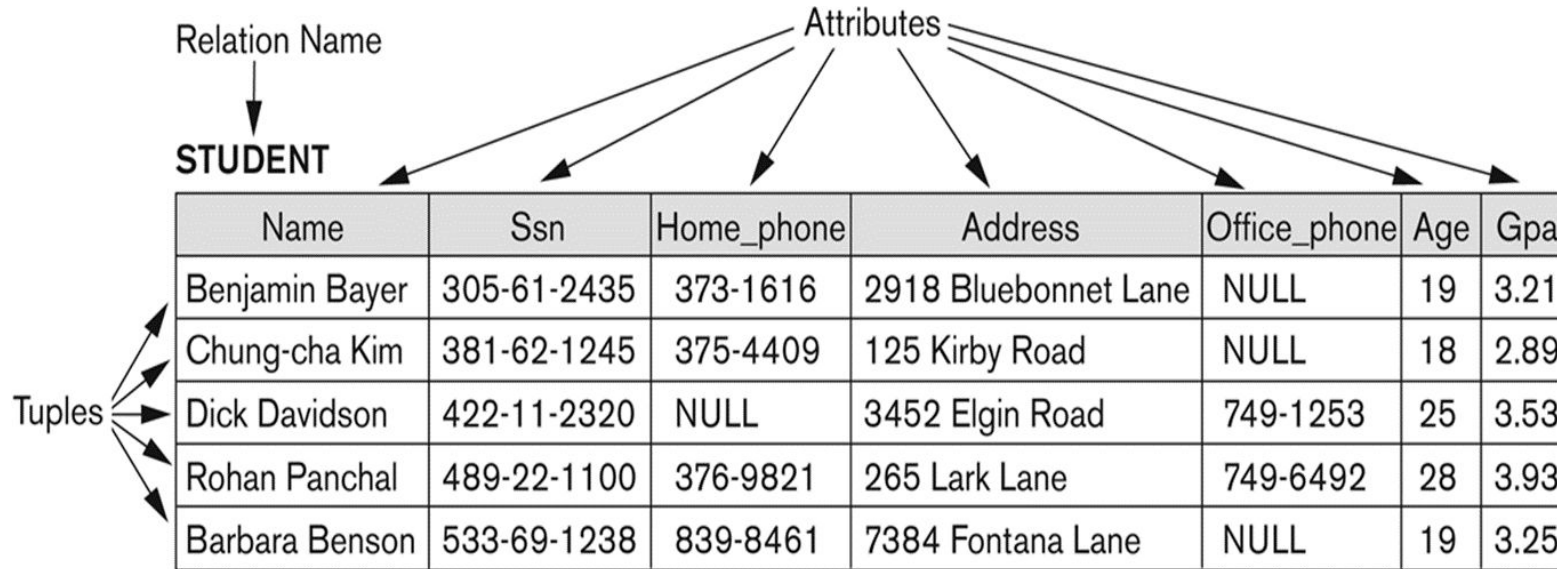


Figure 5.1

The attributes and tuples of a relation STUDENT.

- A **domain** has a logical definition:

Example: “USA_phone_numbers” are the set of 10 digit phone numbers valid in the U.S.

- A domain also has a data-type or a format defined for it.
- The USA_phone_numbers may have a format: (ddd)ddd-dddd where each d is a decimal digit.
- Dates have various formats such as year, month, date formatted as yyyy-mm-dd, or as dd mm,yyyy etc.

The **Schema** (or description) of a Relation

- Denoted by $R(A_1, A_2, \dots, A_n)$
- R is the **name** of the relation
- The **attributes** of the relation are $A_1, A_2, \dots, A_n$

Each attribute has a **domain** or a set of valid values.

For example, the domain of Cust-id is 6 digit numbers.

Example:

**CUSTOMER (Cust-id, Cust-name,
Address, Phone#)**

CUSTOMER is the relation name

Defined over the four attributes:

Cust-id, Cust-name, Address,
Phone#

Relation state:

A relation state (r) of a relation schema $R(A_1, A_2, \dots, A_n)$ is a **set of n-tuples**

$$r = \{t_1, t_2, \dots, t_m\}$$

Relational Schema (R) is also known as Intension and Relational State (r) is also known as Extension.

A **tuple** is an ordered set of values

A relation is a **set** of such tuples (rows)

Key of a Relation:

- Each row has a value of a data item (or set of items) that uniquely identifies that row in the table called the *key*

In the STUDENT relation, SSN is the key

- Sometimes row-ids or sequential numbers are assigned as keys to identify the rows in a table called *artificial key* or *surrogate key*

Constraints determine which values are permissible and which are not in the database. They are of three main types:

1. **Inherent model based or Implicit Constraints:**

Based on the data model itself. (E.g., relational model does not allow a list as a value for any attribute)

2. **Schema-based or Explicit Constraints:**

Expressed in the schema by using the facilities provided by the model. (E.g., max. cardinality ratio constraint in the ER model)

3. **Application based or semantic constraints:**

Beyond the expressive power of the model and must be specified and enforced by the application programs.

There are 5 types of Schema Based Constraints:

- **Domain** constraint
- **Key** constraints
- **Constraints on NULLs**
- **Entity integrity** constraints
- **Referential integrity** constraints

Domain constraint: Every value in a tuple must be from the domain of its attribute (or it could be null, if allowed for that attribute)

Key constraints:

Definitions:

Super key: Is a set of attributes SK of R with the following condition:

- No two tuples in any valid relation state $r(R)$ will have the same value for SK
- That is, for any distinct tuples t_1 and t_2 in $r(R)$, $t_1[SK] \neq t_2[SK]$
- This condition must hold in any valid state $r(R)$

Key: A "minimal" superkey

That is, a key is a superkey K such that removal of any attribute from K results in a set of attributes K' that is not a superkey (does not possess the superkey uniqueness property)

A Key is a Superkey but not vice versa.

Candidate key: A relation schema can have more than one key, each of these keys is called candidate key.

Example: Car relation has more keys (registration number / engine_serial no/license number).

Primary key: If a relation has several candidate keys, one is chosen arbitrarily to be the primary key.

- The primary key attributes are underlined.
- Example: Consider the CAR relation schema:
 - CAR(licenseno, Reg#, SerialNo, Make, Model, Year)
 - We chose SerialNo as the primary key
- The primary key value is used to uniquely identify each tuple in a relation - Provides the tuple identity

Constraints of Nulls:

NULL constraint specifies whether NULL values are permitted or not permitted

Example: if every student tuple must have a valid, non NULL value for the Name attribute, then Name of STUDENT is constrained to be NOT NULL.

Entity Integrity Constraint:

- The primary key attributes PK of each relation schema R in S cannot have null values in any tuple of $r(R)$.
- This is because primary key values are used to identify the individual tuples.
- $t[PK] \neq \text{null}$ for any tuple t in $r(R)$

Referential Integrity Constraint:

- A constraint involving two relations is used to specify a relationship among tuples in two relations the referencing relation and the referenced relation.

Foreign key: A set of attributes FK in relation schema R1 is a foreign key of R1 that references relation R2 if it satisfies the following rules:

1: The attributes in FK have the same domain(s) as the primary key attributes PK of R2; the attributes FK are said to reference or refer to the relation R2.

2: A value of FK in a tuple t_1 of the current state $r_1(R_1)$ either occurs as a value of PK for some tuple t_2 in the current state $r_2(R_2)$ or is NULL. In the former case, we have $t_1[FK]=t_2[PK]$, and we say that the tuple t_1 references or refers to the tuple t_2 .

Referential Integrity Constraint:

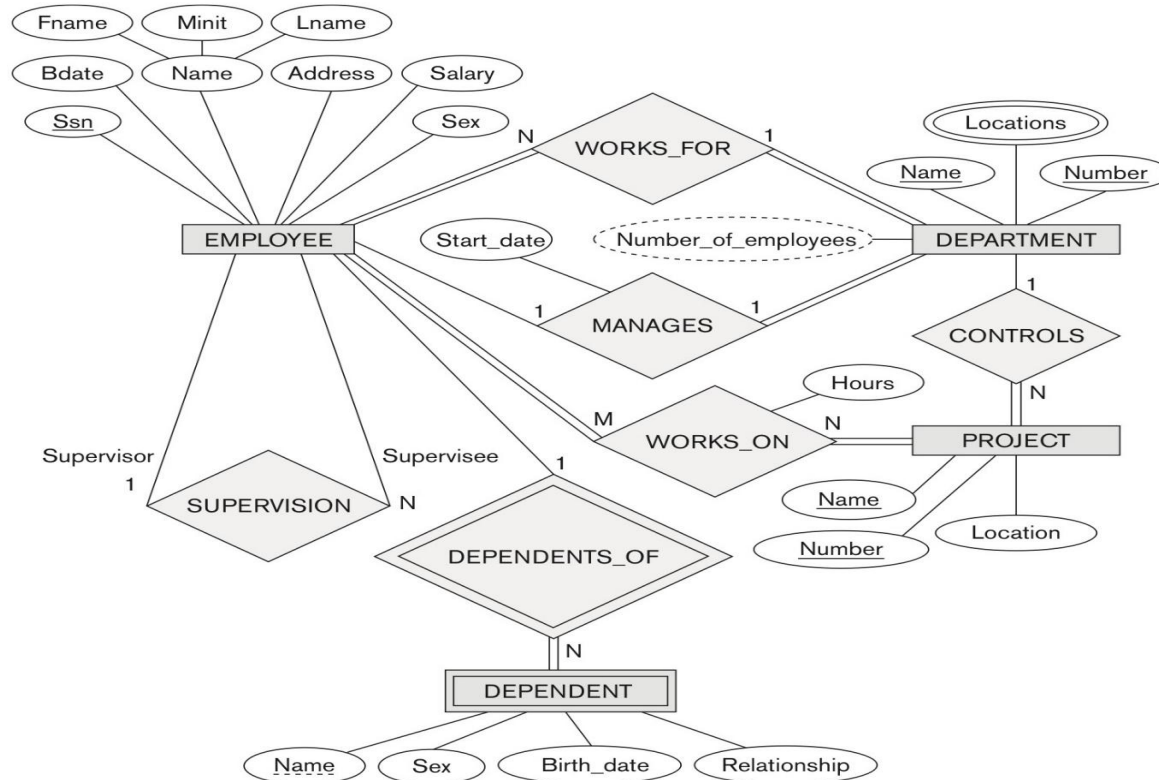
- The value in the foreign key column (or columns) FK of the **referencing relation** R1 can be **either**:
 - (1) a value of an existing primary key value of a corresponding primary key PK in the **referenced relation** R2, or
 - (2) a **null** if allowed

- Both the E-R model and the relational database model are abstract and logical representations of real-world organizations.
- Due to shared design principles, an E-R design can be transformed into a relational design.
- For each entity set and for each relationship set in the database design, there is a unique relation schema to which we assign the name of the corresponding entity set or relationship set.

ER-to-Relational Mapping Algorithm

- Step 1: Mapping of Regular/Strong Entity Types
- Step 2: Mapping of Weak Entity Types
- Step 3: Mapping of Binary 1:1 Relation Types
- Step 4: Mapping of Binary 1:N Relationship Types.
- Step 5: Mapping of Binary M:N Relationship Types.
- Step 6: Mapping of Multivalued attributes.
- Step 7: Mapping of N-ary Relationship Types.

ER Conceptual schema diagram for Company Database



Step 1: Mapping of Regular/Strong Entity Types.

- For each regular (strong) entity type E in the ER schema, create a relation R that includes all the simple attributes of E.
- Choose one of the key attributes of E as the primary key for R.
- If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R.

Example: We create the relations EMPLOYEE, DEPARTMENT, and PROJECT in the relational schema corresponding to the regular entities in the ER diagram.

- SSN, DNUMBER, and PNUMBER are the primary keys for the relations EMPLOYEE, DEPARTMENT, and PROJECT as shown.

Step 2: Mapping of Weak Entity Types

- For each weak entity type W in the ER schema with owner entity type E, create a relation R & include all simple attributes (or simple components of composite attributes) of W as attributes of R.
- Also, include as foreign key attributes of R the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s).
- The primary key of R is the *combination* of the primary key(s) of the owner(s) and the partial key of the weak entity type W, if any.

Example: Create the relation DEPENDENT in this step to correspond to the weak entity type DEPENDENT.

- Include the primary key SSN of the EMPLOYEE relation as a foreign key attribute of DEPENDENT (renamed to ESSN).
- The primary key of the DEPENDENT relation is the combination {ESSN, DEPENDENT_NAME} because DEPENDENT_NAME is the partial key of DEPENDENT.

Step 3: Mapping of Binary 1:1 Relation Types

For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R.

There are three possible approaches:

1: Foreign Key (2 relations) approach: Choose one of the relations-say S-and include a foreign key in S the primary key of T. It is better to choose an entity type with total participation in R in the role of S.

Example: 1:1 relation MANAGES is mapped by choosing the participating entity type DEPARTMENT to serve in the role of S, because its participation in the MANAGES relationship type is total.

Step 3: Mapping of Binary 1:1 Relation Types

- 2. Merged relation (1 relation) option:** An alternate mapping of a 1:1 relationship type is possible by merging the two entity types and the relationship into a single relation. This may be appropriate when both participations are total.
- 3. Cross-reference or relationship relation (3 relations) option:**
The third alternative is to set up a third relation R for the purpose of cross-referencing the primary keys of the two relations S and T representing the entity types.

Step 4: Mapping of Binary 1:N Relationship Types.

- For each regular binary 1:N relationship type R, identify the relation S that represent the participating entity type at the N-side of the relationship type.
- Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R.
- Include any simple attributes of the 1:N relation type as attributes of S.

Step 4: Mapping of Binary 1:N Relationship Types.

Example: 1:N relationship types WORKS_FOR, CONTROLS, and SUPERVISION in the figure.

- For WORKS_FOR we include the primary key DNUMBER of the DEPARTMENT relation as foreign key in the EMPLOYEE relation and call it DNO.

An alternative approach is to use a Relationship relation (cross referencing relation) – this is rarely done.

Step 5: Mapping of Binary M:N Relationship Types

- For each regular binary M:N relationship type R , *create a new relation S* to represent R . This is a *relationship relation*.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types; *their combination will form the primary key* of S .
- Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S .

Step 5: Mapping of Binary M:N Relationship Types

Example: The M:N relationship type WORKS_ON from the ER diagram is mapped by creating a relation WORKS_ON in the relational database schema.

- The primary keys of the PROJECT and EMPLOYEE relations are included as foreign keys in WORKS_ON and renamed PNO and ESSN, respectively.
- Attribute HOURS in WORKS_ON represents the HOURS attribute of the relation type. The primary key of the WORKS_ON relation is the combination of the foreign key attributes {ESSN, PNO}.

Step 6: Mapping of Multivalued attributes.

- For each multivalued attribute A, create a new relation R.
- This relation R will include an attribute corresponding to A, plus the primary key attribute K-as a foreign key in R-of the relation that represents the entity type of relationship type that has A as an attribute.
- The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.

Step 6: Mapping of Multivalued attributes.

Example: The relation DEPT_LOCATIONS is created.

- The attribute DLOCATION represents the multivalued attribute LOCATIONS of DEPARTMENT, while DNUMBER-as foreign key-represents the primary key of the DEPARTMENT relation.
- The primary key of R is the combination of {DNUMBER, DLOCATION}.

Step 7: Mapping of N-ary Relationship Types.

- For each n-ary relationship type R, where $n > 2$, create a new relationship S to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.
- Also include any simple attributes of the n-ary relationship type (or simple components of composite attributes) as attributes of S.

Example: The relationship type SUPPLY in the ER on the next slide.

- This can be mapped to the relation SUPPLY shown in the relational schema, whose primary key is the combination of the three foreign keys {SNAME, PARTNO, PROJNAME}

Result of mapping COMPANY ER schema into a relational database schema

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	------------	-------	---------	-----	--------	-----------	-----

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

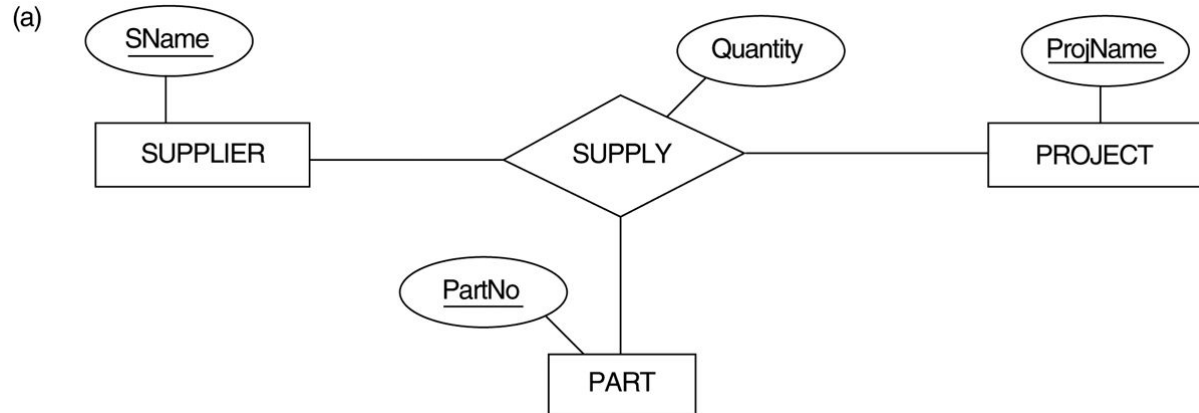
Pname	<u>Pnumber</u>	<u>Plocation</u>	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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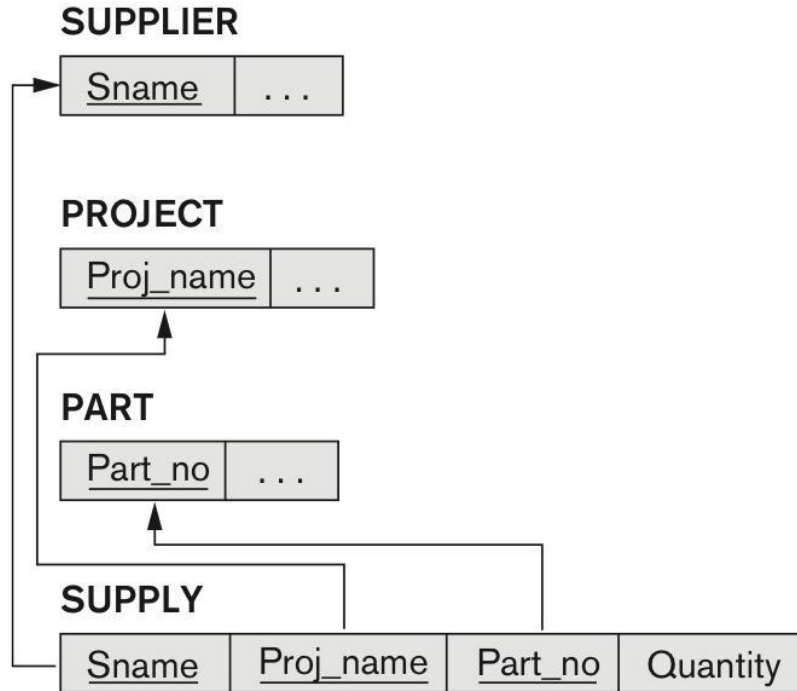


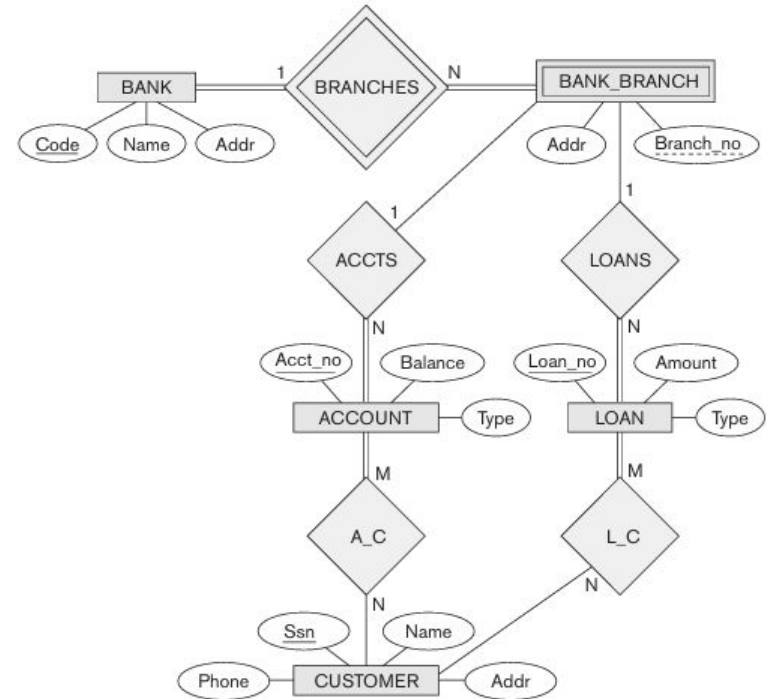
Table 9.1 Correspondence between ER and Relational Models

ER MODEL	RELATIONAL MODEL
Entity type	<i>Entity</i> relation
1:1 or 1:N relationship type	Foreign key (or <i>relationship</i> relation)
M:N relationship type	<i>Relationship</i> relation and <i>two</i> foreign keys
<i>n</i> -ary relationship type	<i>Relationship</i> relation and <i>n</i> foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key

Consider the ER diagram shown in the figure for part of a BANK database.

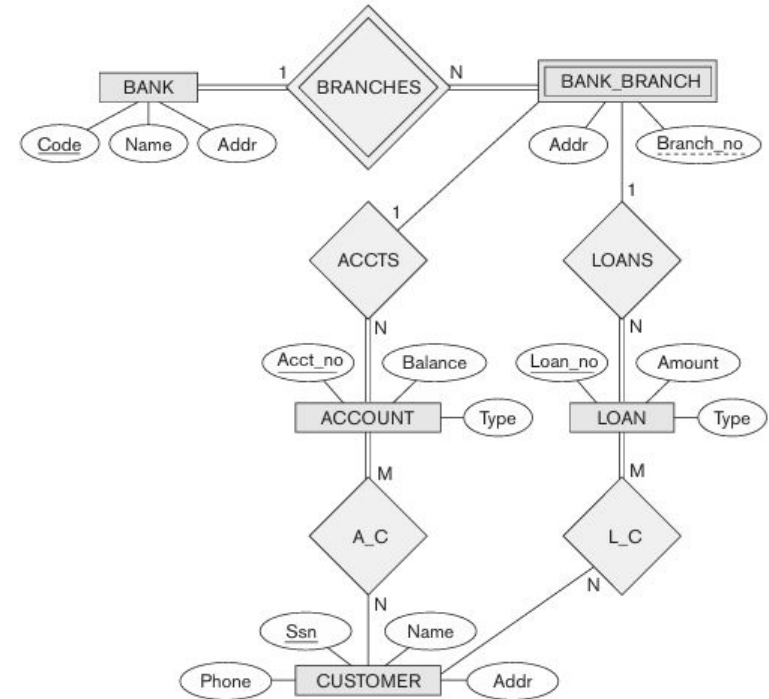
Each bank can have multiple branches, and each branch can have multiple accounts and loans.

- List the strong entity types.
- Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
- What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?



- List the strong entity types.
LOAN, BANK, ACCOUNT and CUSTOMER
- Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
Yes, there is a weak entity **BANK_BRANCH and its identifying relationship is **BRANCHES**. It's partial Key is **Branch_no****

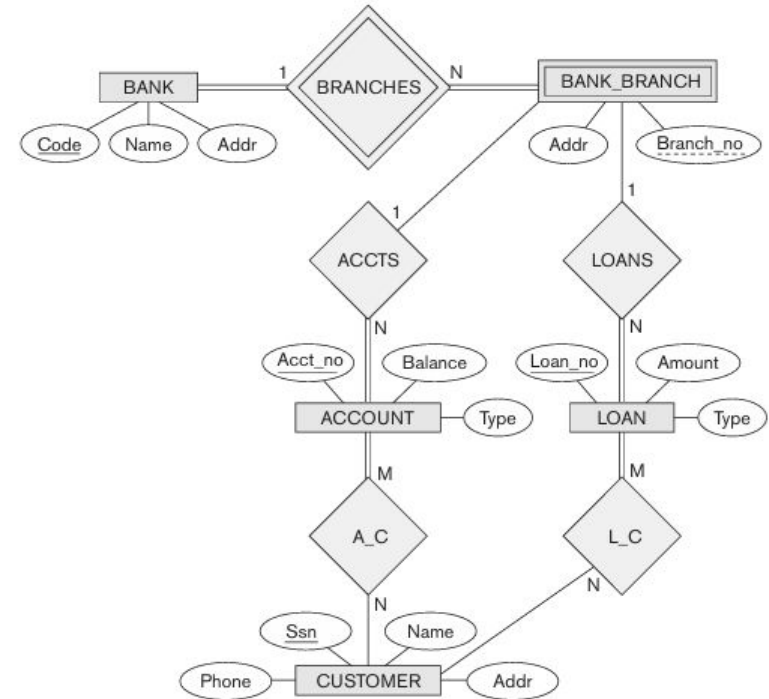
Yes, there is a weak entity **BANK_BRANCH and its identifying relationship is **BRANCHES**. It's partial Key is **Branch_no****



- What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?

The constraints are:

- No two branches have the same number**
- A bank can have any number of branches but a branch only belongs to one bank.**



1. Which of the following is the first step in converting an ER diagram into a relational schema?

- A) Mapping weak entity sets
- B) Mapping strong entity sets
- C) Mapping relationships
- D) Mapping composite attributes

2. What is the primary key in the relational schema for a weak entity set?

- A) The discriminator of the weak entity set
- B) The primary key of the identifying entity set
- C) The combination of the weak entity's discriminator and the identifying entity's primary key
- D) A randomly generated primary key

3. In the relational schema for a multivalued attribute, which constraint is added?

- A) Primary key constraint
- B) Foreign key constraint
- C) Unique constraint
- D) No constraint is added

4. How is a derived attribute handled in a relational schema?

- A) It is stored as a regular attribute
- B) It is represented as a new relation
- C) It is represented using a stored procedure or function
- D) It is merged with other attributes

Answers:

1. B
2. C
3. B
4. C



THANK YOU

Anchor: Dr. Nagasundari S, Dept of CSE

nagasundaris@pes.edu